

### Project Initialization and Planning Phase

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Date          | 07 oct 2025                                                                          |
| Team ID       | xxxxxx                                                                               |
| Project Title | Predicting Plant Growth Stages with Environmental and Management Data Using Power BI |
| Maximum Marks | 3 Marks                                                                              |

### Project Proposal (Proposed Solution)

| Project Overview  |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | To develop a data-driven classification system that accurately predicts and classifies plant growth stages based on environmental and soil parameters such as temperature, humidity, sunlight, and soil nutrients.                                                                                                                                              |
| Scope             | The project focuses on collecting, preprocessing, analyzing, and classifying plant growth data using machine learning techniques. The system will help farmers, agricultural researchers, and policymakers understand the impact of environmental factors on plant health and make informed, data-backed decisions for improving crop yield and sustainability. |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                                                 |
| Description       | Farmers and agricultural researchers struggle to interpret large amounts of environmental data that affect plant growth. Manual analysis is time-consuming, inconsistent, and lacks predictive insights. This makes it difficult to identify which environmental factors most influence plant health and yield.                                                 |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Impact                   | Solving this problem will enable more efficient crop management, optimized resource utilization (water, fertilizer, and sunlight), and improved yield prediction. It will also support sustainable agriculture practices through data-driven decision-making.                                                                                                                                                                               |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Approach                 | he proposed system will utilize machine learning algorithms to classify plant growth stages. Steps include data preprocessing, feature extraction, model training (using classifiers like Decision Tree, Random Forest, or Logistic Regression), and performance evaluation using metrics such as accuracy and F1-score. The results will be visualized using graphs and dashboards for user understanding.                                 |
| Key Features             | <ul style="list-style-type: none"><li>• Data cleaning, normalization, and visualization</li><li>• Machine learning–based growth stage classification</li><li>• Insightful graphical reports showing factor impacts (e.g., sunlight vs. growth)</li><li>• Simple and user-friendly interface for uploading data and viewing predictions</li><li>• Scalable solution for future integration with IoT-based plant monitoring systems</li></ul> |

### Resource Requirements

| Resource Type       | Description                             | Specification/Allocation   |
|---------------------|-----------------------------------------|----------------------------|
| <b>Hardware</b>     |                                         |                            |
| Computing Resources | CPU/GPU specifications, number of cores | e.g., 2 x NVIDIA V100 GPUs |
| Memory              | RAM specifications                      | e.g., 8 GB                 |

|                         |                                       |                                     |
|-------------------------|---------------------------------------|-------------------------------------|
| Storage                 | Disk space for data, models, and logs | e.g., 1 TB SSD                      |
| <b>Software</b>         |                                       |                                     |
| Frameworks              | Python frameworks                     | e.g., Flask                         |
| Libraries               | Additional libraries                  | e.g., scikit-learn, pandas, numpy   |
| Development Environment | IDE, version control                  | e.g., Jupyter Notebook, Git         |
| <b>Data</b>             |                                       |                                     |
| Data                    | Source, size, format                  | e.g., Kaggle dataset, 10,000 images |